PROJECT: INVESTIGATING NETFLIX MOVIES

Netflix! What started in 1997 as a DVD rental service has since exploded into one of the largest entertainment and media companies.

Given the large number of movies and series available on the platform, it is a perfect opportunity to flex your exploratory data analysis skills and dive into the entertainment industry.

You work for a production company that specializes in nostalgic styles. You want to do some research on movies released in the 1990's. You'll delve into Netflix data and perform exploratory data analysis to better understand this awesome movie decade!

You have been supplied with the dataset `netflix_data.csv`, along with the following table detailing the column names and descriptions. Feel free to experiment further after submitting!

The data

netflix_data.csv

Column	Description
`show_id`	The ID of the show
`type`	Type of show
`title`	Title of the show
`director`	Director of the show
`cast`	Cast of the show
`country`	Country of origin
`date_added`	Date added to Netflix
`release_year`	Year of Netflix release
`duration`	Duration of the show in minutes
`description`	Description of the show
`genre`	Show genre

```python
# Importing pandas and matplotlib
import pandas as pd
import matplotlib.pyplot as plt

# Read in the Netflix CSV as a DataFrame
netflix_df = pd.read_csv("netflix_data.csv")
```

```python
# Start coding here! Use as many cells as you like
```

```
netflix_df_1990 = netflix_df[(netflix_df["release_year"] < 2000) & (netflix_df["release_year"]> 1990)]
netflix_df_1990
```

...	...	...	title	director	cast
6	s8	Movie	187	Kevin Reynolds	Samuel L. Jackson, John Hear
118	s167	Movie	A Dangerous Woman	Stephen Gyllenhaal	Debra Winger, Barbara Hersh
145	s211	Movie	A Night at the Roxbury	John Fortenberry	Will Ferrell, Chris Kattan, Dan
167	s239	Movie	A Thin Line Between Love & Hate	Martin Lawrence	Martin Lawrence, Lynn Whitfi
194	s274	Movie	Aashik Awara	Umesh Mehra	Saif Ali Khan, Mamta Kulkarni
315	s456	Movie	American Beauty	Sam Mendes	Kevin Spacey, Annette Bening
320	s467	Movie	American History X	Tony Kaye	Edward Norton, Edward Furlo
333	s487	Movie	An American Tail: Fievel Goes West	Phil Nibbelink, Simon Wells	Philip Glasser, James Stewart
334	s488	Movie	An American Tail: The Mystery of the Night ...	Larry Latham	Thomas Dekker, Lacey Chabe
352	s508	Movie	Andaz Apna Apna	Rajkumar Santoshi	Aamir Khan, Salman Khan, Ra
410	s593	Movie	As Good as It Gets	James L. Brooks	Jack Nicholson, Helen Hunt, G
428	s624	Movie	Auschwitz: Blueprints of Genocide	Mike Rossiter	Roy Marsden
430	s627	Movie	Austin Powers: International Man of Mystery	Jay Roach	Mike Myers, Elizabeth Hurley,
431	s628	Movie	Austin Powers: The Spy Who Shagged Me	Jay Roach	Mike Myers, Heather Graham
468	s688	Movie	Bad Boys	Michael Bay	Will Smith, Martin Lawrence, T
479	s709	Movie	Balto	Simon Wells	Kevin Bacon, Bob Hoskins, Bri

Rows: 169 Expand

```
netflix_df_Movies = netflix_df_1990[netflix_df_1990["type"] == "Movie"]
netflix_df_Movies
```

...	...	...	title	director	cast
6	s8	Movie	187	Kevin Reynolds	Samuel L. Jackson, John Hear
118	s167	Movie	A Dangerous Woman	Stephen Gyllenhaal	Debra Winger, Barbara Hersh
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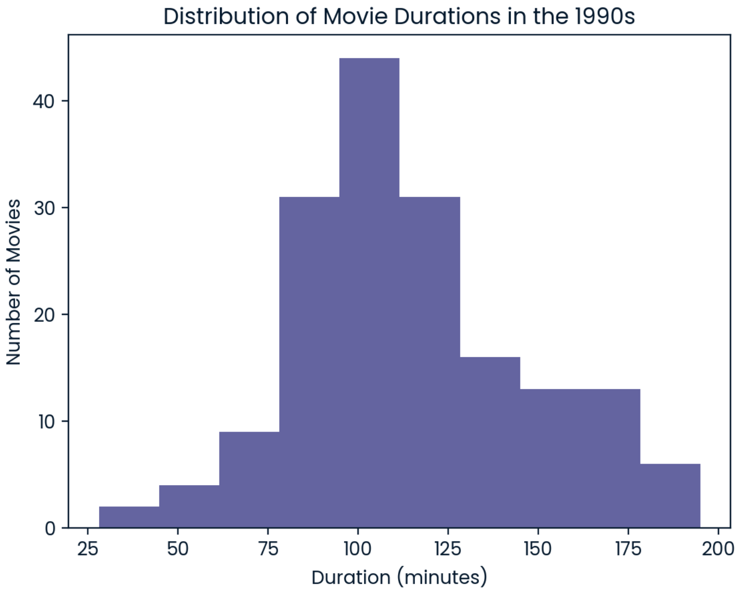
Rows: 169 Expand

```
plt.hist(netflix_df_Movies["duration"])
plt.title('Distribution of Movie Durations in the 1990s')
plt.xlabel('Duration (minutes)')
plt.ylabel('Number of Movies')
plt.show()
```




```
duration = 100
```

```
action_movies = netflix_df_Movies[netflix_df_Movies["genre"] == "Action"]
short_movie_count = 0


for label, row in action_movies.iterrows():
    if row["duration"] < 90:
        short_movie_count = short_movie_count + 1
    else:
        short_movie_count = short_movie_count

print(short_movie_count)
```

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```

```
# Importing pandas and matplotlib
import pandas as pd
import matplotlib.pyplot as plt

# Read in the Netflix CSV as a DataFrame
netflix_df = pd.read_csv("netflix_data.csv")

# Subset the DataFrame for type "Movie"
netflix_subset = netflix_df[netflix_df["type"] == "Movie"]

# Filter the to keep only movies released in the 1990s
# Start by filtering out movies that were released before 1990
subset = netflix_subset[(netflix_subset["release_year"] >= 1990)]

# And then do the same to filter out movies released on or after 2000
movies_1990s = subset[(subset["release_year"] < 2000)]

# Another way to do this step is to use the & operator which allows you to do this type of filtering in one step
# movies_1990s = netflix_subset[(netflix_subset["release_year"] >= 1990) & (netflix_subset["release_year"] < 2000)]

# Visualize the duration column of your filtered data to see the distribution of movie durations
# See which bar is the highest and save the duration value, this doesn't need to be exact!
plt.hist(movies_1990s["duration"])
plt.title('Distribution of Movie Durations in the 1990s')
plt.xlabel('Duration (minutes)')
plt.ylabel('Number of Movies')
plt.show()

duration = 100

# Filter the data again to keep only the Action movies
action_movies_1990s = movies_1990s[movies_1990s["genre"] == "Action"]

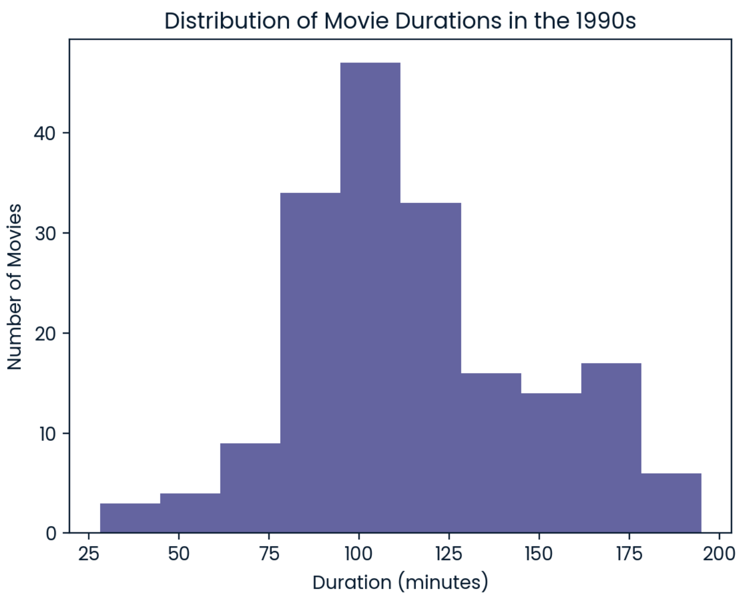
# Use a for loop and a counter to count how many short action movies there were in the 1990s

# Start the counter
short_movie_count = 0

# Iterate over the labels and rows of the DataFrame and check if the duration is less than 90, if it is, add 1 to the counter,
if it isn't, the counter should remain the same
for label, row in action_movies_1990s.iterrows() :
    if row["duration"] < 90 :
        short_movie_count = short_movie_count + 1
    else:
        short_movie_count = short_movie_count

print(short_movie_count)

# A quicker way of counting values in a column is to use .sum() on the desired column
# (action_movies_1990s["duration"] < 90).sum()
```

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```